



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

SECJ 2203: Software Engineering
Semester 01, 2024/2025

PROJECT PROPOSAL
KADA SYSTEM

Team Name: <TechMedico>

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Table of Contents

Item	Page No	Prepared by	Moderated by
1. Introduction	2	CHOH JING YI	
2. Existing Systems	4	LEE YIN SHEN	
3. Proposed System	8	ELIJAH SHE	
4. Software Process Model	10	JOANNE & YUHAN	
5. Project Schedule	15	LEE YIN SHEN	
6. References	16	ALL MEMBERS	

1. Introduction

Farmers in Malaysia especially those live in Kemubu region frequently struggle to raise the money they need to start making money and make investments in new equipment or technology. Since many farmers particularly in developing nations who are lack access to credit or finance that is suitable to their specific needs [1]. However, the KADA system is outdated and finds it difficult to adapt to modern technologies. When it comes to handling registrations, it is ineffective since paper forms remain in use rather than a more efficient digital platform. Users become frustrated as a result of the procedure being slow and easy to make mistakes.

Our group, ABC has decided to create an online registration system to replace the KADA manual registration system. Employees must manually complete a form in order to register as KADA cooperative members under the existing system. This outdated approach is challenging to use and leads to many mistakes and inefficiency. Employees are discouraged from enrolling due to the cumbersome process, as they may find it too complex or inconvenient. Apart from the problems with the registration procedure, the cooperative's existing membership ceiling of 500 members severely restricts the number of employees who are eligible to register. The workforce surpasses the membership cap, which prevents many prospective employees from joining and taking advantage of the cooperative's financial benefits. Before, manual registration wasn't thought of as issues, but as technology advanced, things changed.

In addition, our group determined that the proposed system would be a cross-platform, cloud-based solution that can be accessed via mobile and online apps. The system will automate the registration process, allowing employees to easily sign up as members of the KADA cooperative through an intuitive online form. Employees will no longer need to use paper-based forms to complete the e-registration form, they need to use e registration form to upload the necessary files, and submit their information for approval. After a user registers, the system will automatically monitor their membership status and eligibility for a number of financial incentives including retirement savings programs and loans. Members will also be able to apply for loans, examine and manage their financial contributions, and monitor their payment statuses in real time. By removing the 500-member cap, the system will ease membership growth and allow an unlimited number of employees to sign up.

There are many benefits to our proposed system. First of all, the system provides a fast, efficient, and user-friendly way to register and access the cooperative's financial services, including loans, savings plans, and retirement funds. Employees can sign up for memberships online by automating the registration process, which reduces the possibility of mistakes and eliminates the need for paper-based forms. The registration process will be easier to use and save time, especially for employees who may have previously found it challenging or cumbersome.

Moreover, with the ideology of Industry 4.0, everyone can now access the internet and share or retrieve information online, making the registration process more accessible and in line with modern digital practices [2]. Additionally, the cloud-based platform of the system gives employees more transparency and control over their money by enabling them to apply for loans, manage their membership status, and keep an eye on their financial contributions in real-time. More employees will be able to join the cooperative and take advantage of these financial resources since the maximum 500 member has been removed hence improving their overall financial stability.

Actually, nowadays they are some system or platform that has been created such as Koperasi Rakyat Malaysia (KRM) [3]. However, there are some differences between KRM existing system and our proposed system. For instance, the KRM system uses traditional membership management that member needs to fill form manually, which can be complicated and easy to make mistakes because it depends on human processes. On the other hand, our suggested system makes it easier to use and more accessible by automating the entire registration and membership management process and enabling online registration using a cloud-based, cross-platform solution. In addition, our proposed system provides real-time financial management and tracking, making it simple for employee to keep an eye on their contributions, savings, and loan status which the feature that KRM's platform not fully integrate.

2. Existing System

The cooperative (koperasi) run by Lembaga Kemajuan Pertanian Kemubu (KADA) offers significant benefits to its members, such as farming training, loan application, and lots of agricultural programs. However, there are some critical problems need to be solved in order to make this system more user-friendly and efficient for the users.

The first problem is cumbersome manual registration process. KADA currently offer an entirely manual registration process, whatever during membership registration, loan application or program registration. The users should download the form from the website, then manually fill them, and then submit it to nearby KADA branch. This process often results in wrong information, which will cause inefficient and time-consuming. Moreover, the registration form only available in Malay language, which will give poor user experience for those who are not proficient of the language. As the result, the KADA's reach is restricted, which prevent a diverse group of users from joining and benefiting from its system.

Furthermore, the other problem is the lack of clear instruction of the system. For example, for a new user, they will only see a complicated interface. They is no a clear instruction for user to understand how to become a member, apply for loan, check their registration status and apply for farming program. The lack of guidance will leads to inefficient as the users may not understand the process involved, lead to incomplete application or miss opportunities.

The next problem is the membership cap. The KADA currently has a cap of only 500 member, which limit the number of users who can get benefit from their services. Under this cap, many users unable to register and access the KADA financial benefits like loan. This restriction also limits the growth and development of the KADA organization.

Lastly, one of the main problems is the lacking of financial transparency and reporting. Currently, the KADA does not provide complete month-end or year-end financial reports, include the profit and loss statements to members. As the result, members are unable to track the organization's financial health and understand how their contributions are being managed. This will create a sense of uncertainty and mistrust among members.

These four problems which consists of manual registration, membership cap, unclear website instructions and lack of financial transparency will hinder the KADA ability to serve its

members efficiently and inclusively. By addressing these problems, that will enhance the accessibility and improve the KADA operations and reach.



Figure 1: Main Page of KADA

1 / 5 | 100% +

Borang Daftar Insentifx

Borang Ini Diberi Percuma - Diisi Oleh Pengusaha

BORANG PENDAFTARAN PESAWAH

SKIM BAJA PADI KERAJAAN PERSEKUTUAN (SBPKP)
 INSentif PENGELUARAN PADI - PEMBAJAKAN (IPKP)
 INSentif PENGELUARAN PADI - INPUT PERTANIAN (IIP)
 INSentif PENINGKATAN PENGELUARAN BERAS NEGARA - (IPB)
 KEMENTERIAN PERTANIAN DAN INDUSTRI ASAS TANI MALAYSIA

SBPKP / IPKP / IIP / IPB (1)

moa
 KEMENTERIAN PERTANIAN
 DAN INDUSTRI ASAS TANI

Pertubuhan Peladang Kawasan No. Pendaftaran :
 Tarikh Permohonan :
 No. Kad Pengenalan :

A. MAKLUMAT PENDAFTARAN

MAKLUMAT PENGUSAHA

Nama Pengusaha / Syarikat :
 No. Kad Pengenalan / No. Pendaftaran :
 Alamat Surat Menyurat :
 Poskod : Bandar : No. Tel / HP :

KEAHLIAN PPK *

Ahli PPK / Bukan Ahli PPK : Unit Peladang : No. AHS :

PENERIMAAN BANTUAN

Bil	Perkara	Telah (") Mendapat Bantuan		Jika "Ya" Nyatakan No. Pendaftaran	Keluasan Didaftar (Hektar)
		YA	TIDAK		
1	Subsidi Harga Padi				
2	Skim Baji Padi Kerajaan Persekutuan				
3	Insentif Pengeluaran Padi - Upah Bajak				
4	Insentif Pengeluaran Padi - Input Pertanian				
5	Insentif Peningkatan Pengeluaran Beras				

Figure 2: Manual Form Registration



Figure 3: Google Analytic Missing

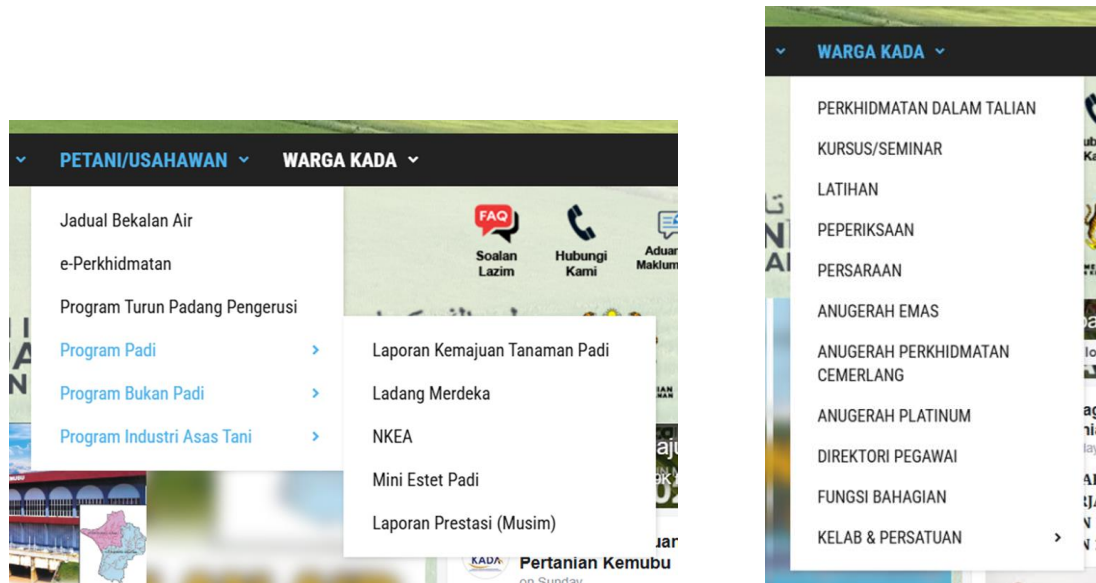


Figure 4: Dazzling Selection

Feature	Lembaga Kemajuan Pertanian Kemubu (KADA)	Lembaga Pertubuhan Peladang (LPP)	Perbadanana Pembangunan Pertanian Negeri Perak (PPPNP)
Online Registration Process	No	Yes	Yes
Multiple System Language	No	No	Yes
Clear Instruction for New Users	No	Yes	Yes
Online Loan/Program Application	No	Yes	Yes
Financial Transparency (Reports)	No	Yes	Yes

3. Proposed System

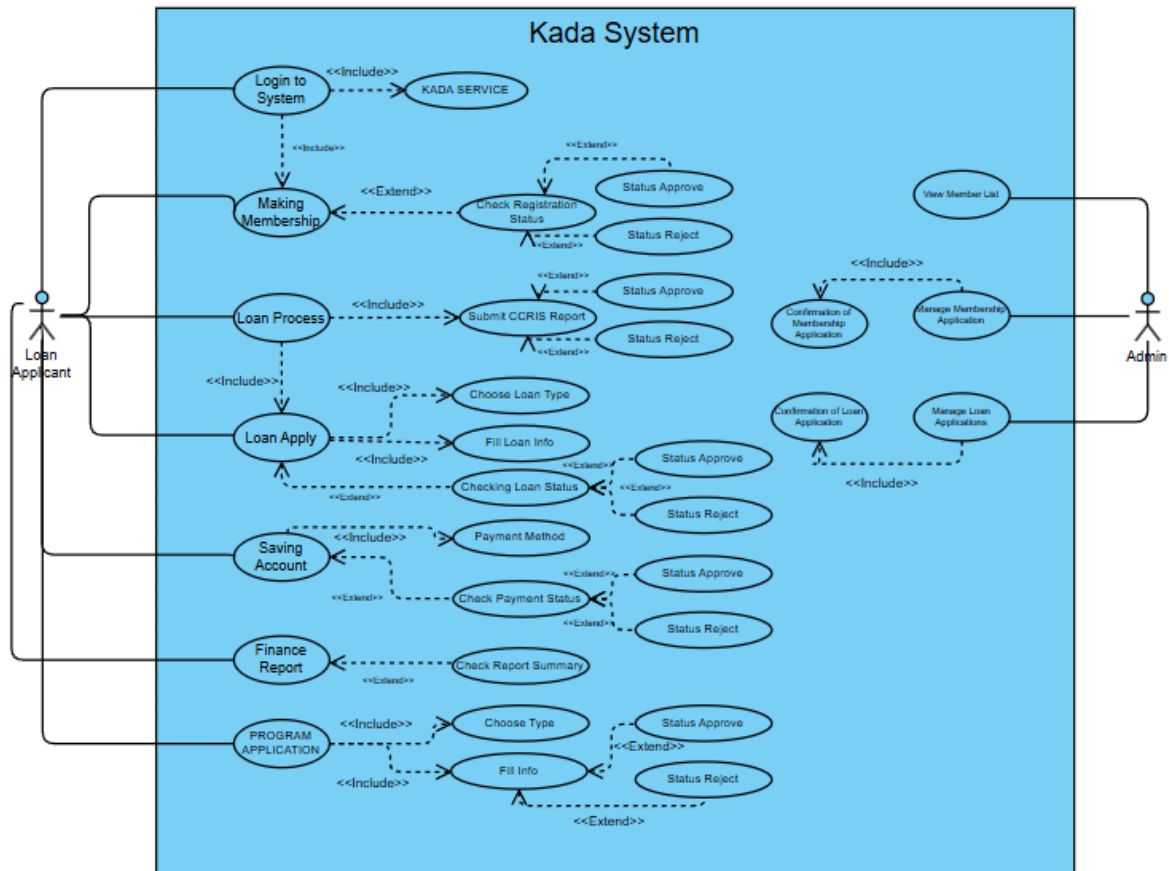


Figure 5: Use Case Diagram for KADA System

The **Kada System** is a comprehensive platform designed to streamline loan application, membership management, saving account handling, finance reporting, and program application processes. The system primarily serves two stakeholder groups: **Loan Applicants** and **Admins**. To access the system, loan applicants must first log in, where the system verifies their credentials and provides relevant information through the KADA service. For new users, the **Membership** process allows applicants to register and become members. After initiating membership registration, applicants can check their registration status, which may lead to either approval or rejection. If rejected, the system notifies applicants of the required corrective actions. Admins manage membership applications by confirming and approving new registrations and maintaining a detailed member list.

Once membership is confirmed, loan applicants can proceed to the **Loan Application** process. This involves **selecting the loan type, and filling out the necessary loan details**. Applicants can track the status of their loan application, which may result in approval or rejection. Approved applicants can proceed with loan repayments, while rejected applicants are informed of the reasons and potential next steps. **Admins manage and oversee the loan approval** process, ensuring that applications comply with the necessary criteria and confirming loan application statuses as needed.

The **Saving Account** feature allows loan applicants to manage their payments related to loans or deposits. They can select their preferred payment method and check the payment status to confirm if the transaction has been processed successfully. The payment status can either be approved or rejected, and the system provides feedback accordingly. Admins oversee these transactions, ensuring accuracy and resolving any discrepancies.

The **Finance Report** feature enables applicants to generate reports summarizing their financial activities, such as loan balances and payment histories. Applicants can check report summaries to keep track of their financial obligations and ensure all transactions are up-to-date. Admins also have access to these reports, which help in monitoring the overall financial activities of the system's users and maintaining compliance with policies.

The **Program Application** process allows applicants to apply for specific financial or administrative programs. This process involves choosing the type of program, filling out the required details, and submitting the application. The application can be approved or rejected based on the provided information. Admins manage these applications, ensuring they are processed correctly and fairly.

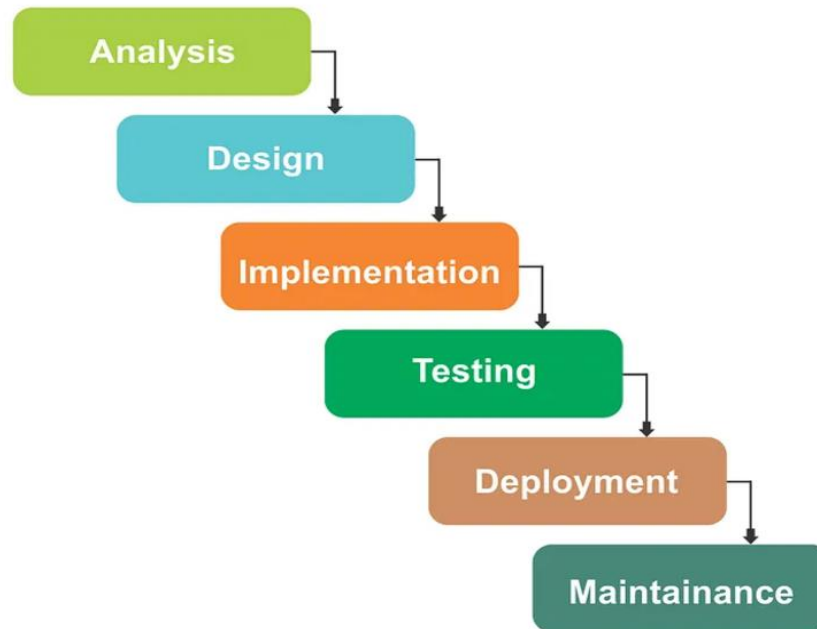
Together, these features ensure that the Kada System offers a seamless, efficient, and transparent platform for managing memberships, loans, saving accounts, finance reports, and program applications. By centralizing these processes, the system enhances user experience and provides a reliable tool for both loan applicants and admins to manage their financial and administrative tasks effectively.

4. Software Process Model

For this project, the Waterfall Model has been selected as the software process model. This methodology works well when there is little client interaction during the development phase and stakeholder requirements are clearly established. We are meeting with stakeholders once throughout the requirement collection phase in accordance with the project requirements. As a result, the Waterfall Model guarantees this project's progress will be linear and orderly.

The Waterfall Model separates the software development life cycle into several phases, including requirement gathering, system design, implementation, testing, deployment and maintenance. As a result, it contains different phases and milestones. Before going on to the following step, each one must be finished in order to guarantee that all development is kept on course. Subsequently, the Waterfall Model permits limited engagement with stakeholders. After the requirement collection phase, stakeholder interaction is minimal because our team and stakeholders are located so far apart that regular meetings are inconvenient. This methodology guarantees that we can lessen our reliance on ongoing feedback. Additionally, the approach places a strong emphasis on documentation at every stage, which guarantees that we fulfill stakeholder requirements and stay in line with them throughout the project's duration. Last but not least, this method is perfect for handling any difficulties that may arise during the construction of a KADA system due to its predictability and simplicity; each step has a clear set of deliverables and a timeframe.

Detailed Activities for SDLC Phases



Requirement gathering

This phase's goal is to gather and record all system requirements from KADA. Conducting a meeting with KADA stakeholders to gain a comprehensive grasp of their present manual system is one of the activities. Next, determine functional requirements such as loan applications and member registration. In order to safeguard members' privacy, we also need to determine non-functional criteria such as security measures. Following this section, we must analyse any bottlenecks or issues that currently exist in the system and then record all of the user and system needs in the Software Requirement Specification (SRS). SRS and user stories or use cases are the deliverables for this phase.

System Design

This goal aims to translate the requirements into a system blueprint. This phase's activities include high-level design, which involves developing system architecture, defining modules, and their relationships. The next step is thorough design, which includes creating a user interface prototype (in this case, we utilize Figma), designing a database schema for efficient data storage, and designing workflows for each process. Furthermore, we must create logical and physical

data flow diagrams. After each phase, the team leader has to decide whether the design meets stakeholder expectations. This phase's deliverables include the design specification document, UI prototype, and database design in ER Diagram.

Implementation

The implementation phase aims to develop the KADA system based on the design specifications created during the previous phase. Activities include breaking the work into development sprints for each module. Module 1 is for member registration and login, module 2 is for loan application processing, and module 3 is for financial reporting. Next, write code for the front-end using HTML, CSS, JavaScript, and Bootstrap. In addition, use a framework such as Laravel to create the backend for database interactions and business logic. After completing these two steps, we can merge both components. Finally, configure a secure database with authentication and role-based control. This phase's deliverables include a functional software prototype for each module as well as a GitHub source code repository.

Testing

This testing phase is important to ensure that the new digital system for Koperasi KADA is reliable, functional, and satisfies all the requirements. This testing phase begins with unit testing, in which every part of the system is tested individually, and all the problems will be recorded in a Unit Test Report. Next, Integration Testing then determines if all modules, such as loan processing and membership registration function together smoothly and produce an Integration Test Report. After that, the complete system is then evaluated by System Testing to make sure it satisfies requirements, and the results are summarized in a System Test Report. Koperasi KADA employees and stakeholders test the system during User Acceptance Testing (UAT) to make sure it satisfies their needs; their comments are recorded in a UAT Report. The system next goes through Debugging and Refining, which fixes any lingering problems and improves performance while creating a Debugging Log.

Deployment

The deployment phases focus on making Koperasi KADA's system into practical use and make sure it is functional. Deployment to Production is the first step, during which the system is configured in the real-world environment. A checklist for the production environment makes sure everything is set up properly. The next step is User Training and Documentation, which trains executives how to use the system efficiently. Video lessons, in-person seminars, and user manuals made using programs like Microsoft Word can all be used as training. User documentation and a training completion report are among the deliverables during this phase. Final Adjustments marks the end of the process, during which any small adjustments are made in response to feedback from the user. To make sure that every modification is noted, this procedure is recorded in an Adjustments Log.

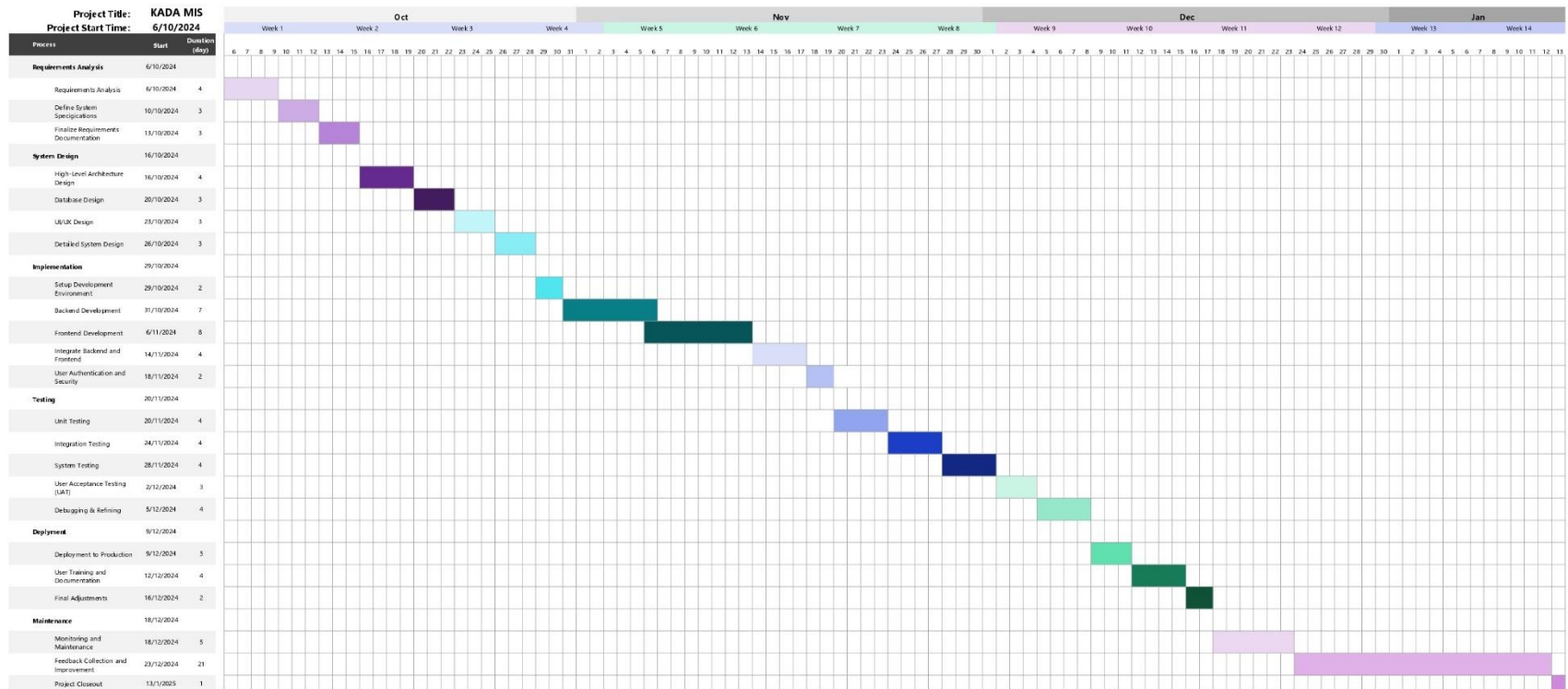
Maintenance

The maintenance phase guarantees Koperasi KADA's system's long-term stability and usability. Monitoring and maintenance involve maintaining tabs on system performance in order to identify and fix all issues. A Maintenance Report summarizes the results of routine actions like backups and updates. In the Feedback Collection and Improvement stage, feedback from users is obtained via questionnaires or meetings. A Feedback Analysis Report and a Future Improvement Plan are the results of using analytics technologies to identify areas that require future improvements. A formal Project Closeout marks the end of the project, during which the system is turned over to Koperasi KADA and a final assessment of its success is carried out. This signifies the successful conclusion of the digital transformation project and is recorded in a Project Closeout Report.

Deliverables and duration

Phase	Duration	Deliverables	Person in-charge
Requirement Analysis	2 weeks	This phase includes requirements analysis report, system specifications document, and finalized requirements documentation.	Joanne, Yu Han
System Design	3 weeks	This phase includes high-level architecture design, database schema, UI/UX mock-ups, and detailed system design documents.	Yin Shen, Jing Yi, Yu Han
Implementation	3 weeks	This phase includes a setup guide for the development environment, backend and frontend codebases, integrated system, and user authentication features.	Elijah, Jing Yi, Joanne
Testing	3 weeks	This phase includes unit testing, integration testing, system testing, UAT feedback, and debugging reports.	Elijah, Yin Shen
Deployment	1 week	This phase includes the deployed system, user training materials, and a final adjustments report.	Yin Shen, Yu Han, Jing Yi
Maintenance	4 weeks	This phase includes maintenance logs, feedback improvement plans, and a project closure report.	Elijah, Joanne, Yin Shen

5. Project Schedule



6. References

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7. Appendices

DATE/TIME	25 Nov 2024 10:00 pm		
LOCATION	M01 KTDI + Online		
AGENDA	1. Understand the details about proposal		
	2. Task distribution for all members		
Meeting MC	ELIJAH SHE YU SHENG		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
LEE YIN SHEN A23CS0236	22:05		
ELIJAH SHE YU SHENG A23CS0073	22:00		
CHOH JING YI A23SC0296	22:00		
JOANNE CHING YIN XUAN A23CS0227	22:05		
LIM YU HAN A23CS0241	22:05		
MINUTES			
NO.	ITEM DISCUSSED	IDEA/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion for proposal	- Elijah shared the rubric and the information of the task - All members read and investigate the task carefully - Clearly identify what is the objective of the task	Elijah
2	Task Distribution	• Jingyi – Introduction • Lee – Existing Systems • Elijah – Proposed System • Joanne & Yuhan – Software Process Model • Lee – Project Schedule • References – All members	All members
3	Meeting Ended		-